

CHEM-1310
Fall 2017
Third Examination
Form A

Multiple Choice - Choose the BEST Answer

Draw the Lewis structure of BrF₅ and use it to answer the next four questions.

1. How many regions of electron density are present on the central atom in BrF₅?

A) 2
B) 3
C) 4
D) 5
E) 6

2. What is the electron geometry of BrF₅?

A) square planar
B) octahedral
C) trigonal bipyramidal
D) seesaw
E) tetrahedral

3. What is the molecular geometry of BrF₅?

A) square pyramidal
B) trigonal pyramidal
C) trigonal bipyramidal
D) seesaw
E) t-shaped

4. The molecular BrF₅ contains which of the following

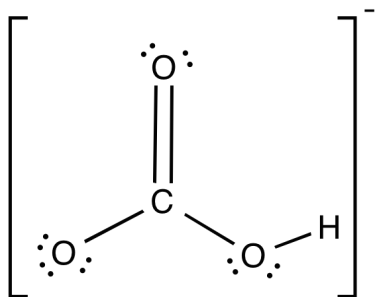
A) ionic bonds and is a polar molecule
B) non-polar covalent bonds and is a nonpolar molecule
C) polar covalent bonds and is a polar molecule
D) nonpolar covalent bonds and is a polar molecule
E) polar covalent bonds and is a nonpolar molecule

Multiple Choice - Choose the BEST Answer

5. The basis of the VSEPR model of molecular bonding is _____.

- A) regions of electron density in the valence shell of an atom will arrange themselves to minimize repulsions
- B) hybrid orbitals will form as necessary to, as closely as possible, achieve spherical symmetry
- C) regions of electron density in the valence shell of an atom will arrange themselves to maximize overlap
- D) regions of electron density on an atom will organize themselves so as to maximize s-character
- E) atomic orbitals of the bonding atoms become hybrid orbitals.

6. Which of the following statements is true regarding the ion shown below?

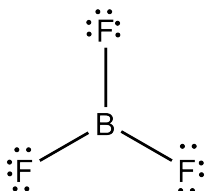


- I. The ion has a resonance structure.
- II. The O-H bond in the ion is the strongest bond present.
- III. There are exactly three σ bonds and one π bond in the ion.

- A) I and II only
- B) I and III only
- C) II only
- D) I only
- E) I, II and III

Multiple Choice - Choose the BEST Answer

7. According to valence bond theory, which orbitals overlap in the formation of the σ B–F bond in the molecular below?

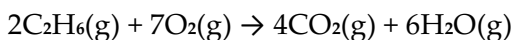


- A) $2p$ on B and $2p$ on F
 - B) sp^2 on B and $2p$ on F
 - C) sp^3 on B and sp^3 on F
 - D) sp on B and sp^3 on F
 - E) The Lewis structure shown above is incorrect; therefore, the orbital formation of the B-F bond cannot be determined.
8. Which of the following correctly explains why we use hybrid orbitals to describe bonding in CH_4 ?
- A) Carbon's s and p orbitals have different energies and are not spatially equivalent.
 - B) Electrons need to be promoted to a higher energy subshell.
 - C) Carbon's orbitals need to be equivalent to hydrogen's orbitals to produce the greatest overlap.
 - D) Hydrogen's s orbitals are too high in energy to form a bond with carbon's p orbitals.
 - E) None of the above are true statements explaining why we use hybrid orbitals to describe bonding in CH_4 .
9. You have a pure sample of $\text{CH}_3\text{OH}(\text{l})$, which of the following lists all of the intermolecular interactions that occur in the sample.

- A) hydrogen bonding, ion-dipole and London dispersion interactions
- B) hydrogen bonding and dipole-dipole interactions
- C) hydrogen bonding, dipole-dipole and London dispersion interactions
- D) dipole-dipole, ion-dipole and London dispersion interactions
- E) dipole-dipole and London dispersion interactions

Multiple Choice - Choose the BEST Answer

10. What is the enthalpy of combustion for ethane, C_2H_6 , as described by the following equation, in kJ per mol of reaction?



Use the information below as needed.

Average Bond Dissociation Energies (kJ/mol)

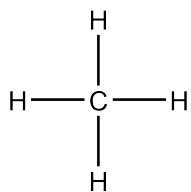
H-C	414
H-O	464
C-O	360
C=O	736
O=O	498
C-C	347
C=C	611

- A) -2.31×10^3
B) -6.65×10^2
C) 3.51×10^2
D) 288
E) There is not enough information.
11. When a soluble ionic compound is placed in water, which of the following occurs?
- A) The ionic compound dissociates into its ions.
B) Hydrogen bonding occurs between the water molecules present.
C) Ion-dipole interaction occurs between the water molecules and the ions present.
D) Hydration occurs.
E) All of the above occur.
12. Which of the following molecules have the highest vapor pressure?
- A) NH_3
B) H_2O
C) CH_4
D) HCl
E) CH_2Cl_2

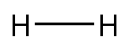
Multiple Choice - Choose the BEST Answer

13. Which of the following molecules is the most polarizable?

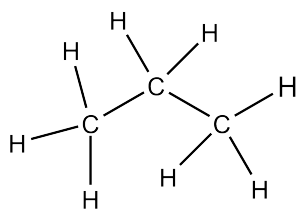
A)



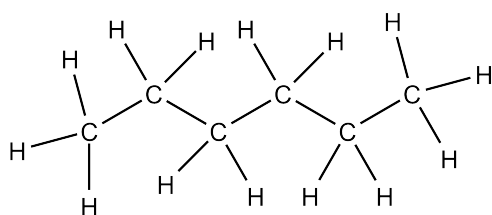
B)



C)



D)



E) All are equally polarizable.

Multiple Choice - Choose the BEST Answer

14. Which process releases the greatest amount of heat to the surroundings?

$$\Delta H_{\text{vap}} = 40.7 \text{ kJ/mol}$$

$$\Delta H_{\text{fus}} = 6.02 \text{ kJ/mol}$$

- A) the condensation of 10.0 g of gaseous H_2O
- B) the freezing of 10.0 g of liquid H_2O
- C) the boiling of 10.0 g of liquid H_2O
- D) the melting of 10.0 g of solid H_2O
- E) the sublimation of 10.0 g of solid H_2O

15. How much energy (in kJ) is required to completely condense 200.0 g of 125.00°C steam?

Use the information below as needed.

Properties of Water

Specific Heats:

$$\text{gas} = 1.84 \text{ J/g}^\circ\text{C}$$

$$\text{liquid} = 4.184 \text{ J/g}^\circ\text{C}$$

$$\text{solid} = 2.09 \text{ J/g}^\circ\text{C}$$

$$\text{Enthalpy of Vaporization: } \Delta H_{\text{vap}} = 40.7 \text{ kJ/mol}$$

$$\text{Enthalpy of fusion: } \Delta H_{\text{fus}} = 6.02 \text{ kJ/mol}$$

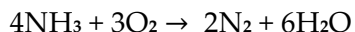
- A) -452 kJ
- B) -406 kJ
- C) -461 kJ
- D) +20.92 kJ
- E) +46 kJ

16. A first-order reaction is 38.5% complete in 520 s. What is the value of the rate constant?

- A) $1.83 \times 10^{-3} \text{ s}^{-1}$
- B) $9.35 \times 10^{-4} \text{ s}^{-1}$
- C) $3.07 \times 10^{-3} \text{ s}^{-1}$
- D) $1.18 \times 10^{-3} \text{ s}^{-1}$
- E) $1.20 \times 10^{-3} \text{ s}^{-1}$

Multiple Choice - Choose the BEST Answer

17. For the following reaction it was found that the rate of formation of N_2 was 0.27 M/s .



At what rate was ammonia being consumed?

- A) -0.27 M/s
- B) 0.54 M/s
- C) 0.81 M/s
- D) 0.13 M/s
- E) 0.77 M/s

18. For the reaction:



a set of experiments are performed, with different initial conditions at the same temperature; the data is shown below. What is the rate law for the reaction?

Experiment	$[\text{BrO}_3^-] \text{ (M)}$	$[\text{Br}^-] \text{ (M)}$	$[\text{H}^+] \text{ (M)}$	Initial Rate (M/s)
1	0.10	0.10	0.10	8.0×10^{-4}
2	0.20	0.10	0.10	1.6×10^{-3}
3	0.20	0.20	0.10	3.2×10^{-3}
4	0.10	0.10	0.20	3.2×10^{-3}

- A) $\text{Rate} = k[\text{Br}^-][\text{H}^+]^2$
- B) $\text{Rate} = k[\text{BrO}_3^-][\text{Br}^-][\text{H}^+]$
- C) $\text{Rate} = k[\text{BrO}_3^-][\text{H}^+]$
- D) $\text{Rate} = k[\text{H}^+]$
- E) $\text{Rate} = k[\text{BrO}_3^-][\text{Br}^-][\text{H}^+]^2$

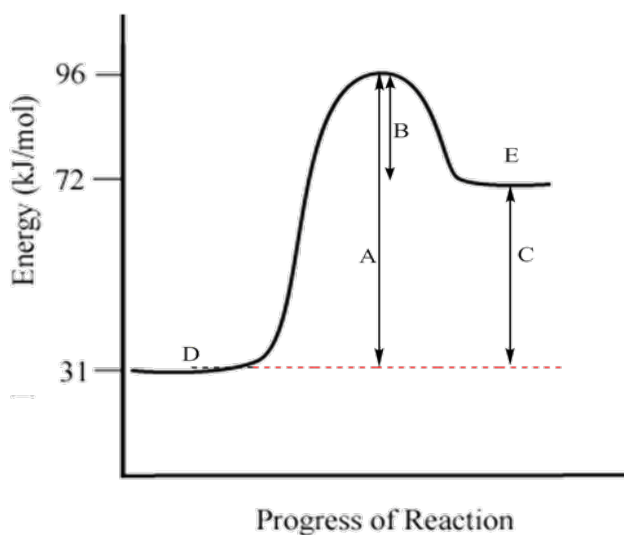
Multiple Choice - Choose the BEST Answer

19. For two molecules to react when they collide, which of the following must occur?

- I. proper orientation of molecules
- II. exothermic reaction
- III. sufficient kinetic energy

- A) I and II only
- B) I and III only
- C) II only
- D) I only
- E) I, II and III

20. Which of the following is not a true statement about the graph below?



- A) The reaction is endothermic.
- B) Line 'C' represents the enthalpy of reaction.
- C) Line 'A' represents the activation energy of the reaction
- D) The transition state of the reaction occurs at the top of the potential energy curve.
- E) All of the above statements are correct.

21. Which form of the exam do you have?

- A) Form A
- B) Form B